

Composing Learners

Populations as namespaces, parallel composition as the operator,
and the ecological relations that typed overlap generates

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Abstract

Two companion notes model learning as the loop the scientific method describes, run by a computation with a finite and conserved token supply in an environment made of other computations [1], and instantiate that model on a solved adversarial game so that its claims can be checked against answers the reader already holds [2]. In both, the learner is an individual and the population is what rescues it: confinement says an individual's incremental revisions cannot leave the ball its early commitments placed it in, and recombination across a lineage is the only escape.

This note takes the population as the learner. A learner is a distribution over a population of scientists, of mortal computations holding no hypothesis, and of resource reservoirs; because that population is carried by a family of names manufactured by quotation, a learner *is* a namespace, described by a name predicate rather than enumerated. Learners then compose by parallel composition, and the whole content of the composition is what their namespaces share.

Four things follow. Disjointness of namespaces is exactly factorisation of the distribution, so composition is strictly monoidal at zero grade and the lax structure map elsewhere *is* the interaction; the deviation from strictness is therefore a measurement rather than a caveat. A single grade is too coarse: the namespace of a learner is typed—metabolic, source, probe, mating—and grading separately over each recovers the classical table of ecological relations as sign conditions rather than importing it. Conservation makes cooperation and competition structurally unlike: tokens are conserved and formulae are not, so mutualism is available only over the probe namespace and competition only over the metabolic and source ones. And disjointness is not stable, because names manufactured independently may coincide; modular composition is purchasable, and the price is a naming discipline that roots each learner's namespace at its own quotation.

We then measure. A second environment—Nim—is ladderred exactly by the same method and schedule as [2], and it inverts that note's headline: in Nim(1, 2, 3) marginal return *rises* up the ladder and the complete theory is the best purchase on it, where in noughts-and-crosses returns collapse and the complete theory is dominated. Whether truth is affordable is a property of the environment, not of the learner, and composition puts several such answers under one conserved budget. Composing the two over a single rate-limited reservoir gives a closed form: each learner retains exactly the fraction $c_{\text{other}}/(c_{\text{self}} + c_{\text{other}})$ of the inquiry it could afford alone. The dilution is regressive—it costs the unfinished science 66.9% of its affordable depth and the finished one nothing—and it is delivered without either learner perceiving, modelling, or touching the other. Finally, exhaustion of a shared reservoir forces the grading vector to change, and only the learner that was confined retains the apparatus to change it. We close on the observation that composition is the exit from the heat death of [2]: a population that has solved its science stops eating, and the cheapest way to make it eat again is to compose it with a learner whose namespace it has not solved.

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1 Introduction

1.1 Why the unit moves up

The mortal scientist of [1] is an individual: a term with a metabolic channel, a hypothesis in guard position, and a walk on the relaxation lattice. Twice in that note the individual turns out to be the wrong unit. Corollary 31 there observes that destructive assay leaves an individual-level hypothesis worthless after its own first test, so the *subject* of a hypothesis must be a namespace rather than a specimen. And Proposition 13 observes that an ultrametric walk of short steps never leaves the ball it began in, so the *holder* of a hypothesis cannot repair a shallow commitment; §12 of that note supplies recombination, and §17.2 states plainly that a population is required rather than preferred.

The worked example makes both visible on numbers [2, §9]. The corner-opening mutant that invades the resident population differs from it by a revision of distance 2^{-1} , is unreachable to any individual by deepening, and is free to a lineage because the cost falls on an offspring holding no confirmations to invalidate.

So the individual is already not the unit of learning; the lineage is. This note asks what happens when the lineage is not the unit either.

1.2 The move, stated once

A learner is a distribution over a population: scientists, mortal computations holding no hypothesis, and resource reservoirs. That population is carried by names, and by Proposition 2 of [2] those names are manufactured by quotation rather than drawn from a stock of atoms, so the family is generated by a decidable predicate and can be *described*. Hence a learner is a namespace, in the sense the hypothesis language already uses for the subject of a hypothesis.

Learners compose by parallel composition, because in this calculus there is nothing else for composition to be. When the namespaces are disjoint the composed learners have non-interacting capabilities. When they overlap they can interact internally, and that interaction includes both cooperation and competition.

Remark 1 (The knower and the known become one kind of thing). This is the observation we would lead with. Corollary 31 of [1] forces the *object* of inquiry to be a namespace; the present move makes the *subject* of inquiry one as well. The two sides of the cut are then described in the same vocabulary, by the same name predicates, at the same prices. That is not tidiness for its own sake: it is what makes a composed learner able to hold a hypothesis about its own components, and what makes the distinction between an internal interaction and an external one a matter of where one draws the cut rather than a matter of sort. Remark 40 of [1] said trophic type is a fact about the cut. The same is now true of learnerhood.

1.3 What we are claiming, and what we are not

We claim that composition of learners is parallel composition on terms, union on namespaces, and *undetermined* on distributions; that the determination of the third is exactly the content of the interaction, and that it factors precisely when the namespaces are disjoint; that a single grade of separation is too coarse and a typed vector of grades recovers the ecological relations rather than importing them; that under conservation cooperation and competition are structurally different rather than opposite in sign; that non-interference is an observation and not a guarantee unless purchased; and that a composite's phase is not a function of its parts' phases, which we exhibit on exact numbers.

We do not claim to have proved a monoidal-category theorem: §4 states the comparison map and the conditions for it to be invertible, and leaves the coherence to the open problems.

We do not claim the access-rate model of §10 is canonical; it is one modelling choice, flagged as such, and the alternative is worked. As in [2], guards below are written in the ideal logic and the current interpreter accepts only the arithmetic and pattern fragment of the **where** clause; occurrences that specify rather than perform a check are marked.

2 What is imported

We recapitulate only what the argument uses, and refer to [1, 2, 3] for the rest.

The cut, and three grades of access. A world is $W \equiv S \mid E$. Computations are ontologically isolated, so E is other computations, and what S can come to know of E is bounded above by E 's context-labelled bisimulation class. Access comes in three grades and no more: *perception*, structural predicates evaluated at the surface, priced by a schedule $\kappa_{\text{sense}}(d)$ monotone in formula depth; *interaction*, the context-labelled modalities $\langle K_j \rangle$, costing a rendezvous per step; and *reflection*, the name predicate $@\varphi$ applied to names S already holds, cheap because it passes through no surface.

The logic is generated, and there is no restriction. Feeding the presentation of the rho calculus to the OSLF family yields one structural connective per term former, one context-labelled modality per rewrite rule and redex position, and—because $|$ carries an associative-commutative theory—a separating conjunction. A name is a quoted process, so $n \models @\varphi \iff n = @P$ with $P \models \varphi$: this is namespace logic [8], in which one describes a namespace by a property rather than enumerating it. Following [2, §2] we treat **new** as sugar and drop the hidden-name quantifier: freshness is manufactured by quotation, and graded separation is graded over a *described namespace* rather than over a restriction vector,

$$P \models \varphi \mid_k^{\mathcal{N}} \psi \iff P \equiv Q \mid R, Q \models \varphi, R \models \psi, \mid \text{fn}(Q) \cap \text{fn}(R) \cap \mathcal{N} \mid = k.$$

Everything below that a modal logic could not do is done with $\mid_k^{\mathcal{N}}$, and §5 turns on the choice of $\mathcal{N}$.

Proposition 2 (Freshness by quotation, [2, Prop. 2]). *Let P be a process and $C[-]$ any context with $C[P] \neq P$. Then $@C[P] \notin n(P)$. Hence a name fresh in P is manufactured by quoting any term properly containing P , with no registry and no consultation.*

Conservation and the game. Tokens are conserved: $\sigma + \kappa = \sigma_0 = \Theta$, with σ held in live stacks and κ migrated into the record. There is no primary producer. Under the internalising encoding $\llbracket - \rrbracket : \mathcal{C}(\text{rho}) \rightarrow \text{rho}$ a located stack becomes a message at a metabolic channel m_P , and harvest is an ordinary receipt on m_P performed by someone else; the predatory contexts are exactly the non-image contexts holding a metabolic name [1, Thm. 25]. A *game* is a choice of admitted context class $\mathcal{A}$ with $\llbracket \mathcal{C}(\text{rho})\text{-Ctx} \rrbracket \subseteq \mathcal{A} \subseteq \text{rho-Ctx}$.

Sources and prey. The trophic distinction is an access profile, not a sort [1, §11.2]: a *source* is a persistent emitter, rate-limited by being a single contract, non-exclusive, non-adaptive, and exhaustible; a *prey* is a linear send, exhaustible, exclusive, and adaptive. By Proposition 41 there, a source-feeder's inquiry converges and a prey-feeder's never does.

The hypothesis space, and confinement. Beneath the characteristic formula χ_P sits a lattice of progressively weaker formulae obtained by forgetting conjuncts; revision is a walk on it, the metric is the ν -stratified ultrametric $d(\varphi, \psi) = 2^{-n}$, and distance, modal depth, and cost are one graded structure. Confinement (Proposition 13 of [1]) says a walk whose steps are all shorter than 2^{-n} never leaves the ball of radius 2^{-n} it began in; cost-monotonicity (Proposition 15) says the escaping move is the expensive one; and bounded branching (Proposition 18) with its corollary *poverty is confining* says the number of siblings a scientist may evaluate is its stack divided by the cost of checking one.

Ecologies as a fibration. A configuration is a partition of Θ across loci together with a structural assignment; an ecology is a configuration with a distribution over reachable configurations. The assignment of ecologies to theories is a fibration rather than a functor, with a canonical section given by taking cost as the weight map, and functoriality requires logic-preserving morphisms because the senses are structural [1, §15]. The present note is largely an argument that this fibration wants a tensor.

3 The learner

3.1 Three layers, and the distribution is one of them

What we have been calling a learner has three components, and it is worth separating them before asking what composition does, because composition acts differently on each.

- (i) **A population.** A term: a parallel composition of scientists, of mortal computations holding no hypothesis, and of reservoirs. This is an object of the calculus.
- (ii) **A namespace.** The family of names the population carries—metabolic channels, source channels, probe channels, mating channels—described by a name predicate rather than listed, which by Proposition 2 is available because the names are manufactured from material the population already holds.
- (iii) **A distribution.** A weighting on the configurations reachable from the population, in the sense of [1, §13]: the microcanonical ensemble at fixed Θ , sampled by the construction of [7].

We take the third to be part of the learner rather than a decoration on it. The alternative—a learner is the population, and the distribution is a decoration indexed over it—would put composition in the base of the fibration of [1, §15], where it is uninformative: two populations composed in the base are just two populations. Putting the distribution inside the learner puts composition in the total space, which is where the interaction lives.

Definition 3 (Learner). A learner is a triple $\mathcal{L} = (P, \mathcal{N}, \mu)$ with P a **rho** term, $\mathcal{N}$ a name predicate satisfied by exactly the names P carries in the sense of Definition 11 below, and μ a distribution over the configurations reachable from P at total $\Theta_{\mathcal{L}} = \sum \sigma_i(0)$. We write $\mathcal{N}(\mathcal{L})$ and $\mu(\mathcal{L})$ for the components.

Remark 4 (The population is not a bag of learners). Definition 3 does not say the population is a set of individual learners composed somehow. It says the population, weighted, *is* the learner, and that individual scientists inside it are components with no special status—exactly as a lineage in [1, §12.10] is a learner without any of its members being the learner. The nesting is real and the note uses it: §12 shows the three units of learning are distinguished by nothing but the revision distance each can reach.

4 Composition

4.1 One operator, three actions

Definition 5 (Composition). For learners $\mathcal{L}_1 = (P_1, \mathcal{N}_1, \mu_1)$ and $\mathcal{L}_2 = (P_2, \mathcal{N}_2, \mu_2)$,

$$\mathcal{L}_1 \mid \mathcal{L}_2 := (P_1 \mid P_2, \mathcal{N}_1 \vee \mathcal{N}_2, \mu_{12}), \quad \Theta_{\mathcal{L}_1 \mid \mathcal{L}_2} = \Theta_{\mathcal{L}_1} + \Theta_{\mathcal{L}_2},$$

where $\mathcal{N}_1 \vee \mathcal{N}_2$ is the disjunction of the two name predicates and μ_{12} is the distribution over configurations reachable from $P_1 \mid P_2$.

On the first component the operator is the calculus' own parallel composition: associative and commutative up to structural congruence, free, and adding nothing. On the second it is disjunction of predicates, which is again free, and is where describing rather than enumerating pays—one does not have to compute a union of two sets of names, one takes an $\vee$ of two formulae. On the third the operator is not determined by its arguments at all. That is the entire subject of this section.

Proposition 6 (Conservation composes, and only conservation does). *Θ is additive under composition, and it is the only quantity in [1, §13] that is. The order parameters are not: ι , the fraction of Θ internal to scientists, is a stack-weighted mean of the parts' values, and δ , the mean concealment depth of external stacks, is not a function of the parts' values at all, because each learner's members become candidate subjects for the other and enter the other's mean.*

Proof sketch. Additivity of Θ is immediate from the ledger law and the fact that composition mints nothing. For ι , write $\iota_i = \sigma_i^{\text{sci}}/\Theta_i$; then $\iota_{12} = (\sigma_1^{\text{sci}} + \sigma_2^{\text{sci}})/(\Theta_1 + \Theta_2)$, the weighted mean. For δ , the mean is taken over the external stacks *available to* a learner, and P_2 's stacks are external to P_1 and were not in P_1 's index set. So δ_{12} depends on the depth at which each learner's stacks lie relative to the other's senses, a quantity computed from the pair and not held by either. $\square$

That δ is a property of the pair is the first sign that composition is not going to be innocent, and §9 makes it a measurement.

4.2 Disjointness is factorisation

Theorem 7 (Factorisation). *Let $\mathcal{L}_1, \mathcal{L}_2$ be learners and let $\mathcal{N} = \mathcal{N}_1 \vee \mathcal{N}_2$. The following are equivalent.*

- (1) $P_1 \mid P_2 \models \top \mid_0^{\mathcal{N}} \top$ with the split at P_1, P_2 : the two namespaces share no name.
- (2) No reduction of $P_1 \mid P_2$ has its redex spanning the two parands.
- (3) $\mu_{12} = \mu_1 \otimes \mu_2$: the composite's distribution is the product of the parts'.

Proof sketch. (1) $\Rightarrow$ (2): the rho calculus has one interaction rule, and it requires a send and a receipt on structurally equal names. A redex spanning the parands therefore witnesses a name in $\text{fn}(P_1) \cap \text{fn}(P_2) \cap \mathcal{N}$, contradicting $k = 0$. (2) $\Rightarrow$ (3): with no spanning redex, every step of $P_1 \mid P_2$ is a step of one parand in a context that does not participate, so the reachable configurations are pairs of independently reachable configurations, and the weight of a pair is the product of the weights since the cost of a step is charged to the parand taking it and the ledger is additive. (3) $\Rightarrow$ (1): if a name is shared, some configuration in which the corresponding rendezvous has fired has weight not equal to the product of its marginals—the two parands' stacks after such a step are correlated by the transfer. $\square$

Remark 8 (Why this is the right statement). Theorem 7 is what makes “the namespaces are disjoint” worth saying. Disjointness of *names* is a syntactic condition; independence of *distributions* is the thing one actually wants when composing learners, because it is what licenses reasoning about each in isolation. The theorem says the syntactic condition is exactly the probabilistic one, and the graded separation $|_k^{\mathcal{N}}$ measures the gap when it fails. We know of no way to say this without a spatial layer: a purely behavioural logic cannot state the split at all, which is [3, Rem. 5] arriving in a new place.

4.3 The lax structure map is the interaction

Composition of learners is a candidate monoidal structure on the total space of the fibration $\pi : \text{Eco} \rightarrow \text{GSLT}_{\mathcal{C}}$ of [1, §15], with unit the empty learner $(\mathbf{0}, \perp, \delta_{\emptyset})$. The obvious question is whether the canonical section s —cost as the weight map—is monoidal, and the answer is no, in a way we think is more useful than a yes.

Proposition 9 (Laxity, and what its defect measures). *There is a comparison morphism*

$$\lambda_{\mathcal{L}_1, \mathcal{L}_2} : s(\mathcal{L}_1) \otimes s(\mathcal{L}_2) \longrightarrow s(\mathcal{L}_1 \mid \mathcal{L}_2)$$

sending a pair of independently sampled trajectories to the interleaving in which no spanning redex fires. It is an isomorphism if and only if the grade is zero, and in general it is not epi: the configurations in the image are exactly those reachable without interaction. The deviation—the mass μ_{12} assigns outside the image of λ —is the probability that the composite does something neither part could do.

Proof sketch. Existence: the non-interacting interleavings of two independent trajectories are reachable in the composite, and their weights are the products, since costs are additive. Iso at $k = 0$: Theorem 7. Failure of surjectivity at $k > 0$: a spanning redex is enabled and fires with positive rate under any weight map assigning positive weight to enabled steps, and the resulting configuration is not a product of independently reachable ones. $\square$

Remark 10 (The good news is that laxity is a number). Conjecture 82 of [1] guesses that the induced map on distributions is only lax and that the natural target is distributions ordered by stochastic dominance. Composition sharpens the guess into something one can compute: the defect of λ is the total interaction, and $|_k^{\mathcal{N}}$ resolves it by which names carry it. Cooperation and competition are then not two phenomena to be modelled but two *signs* of one measured quantity, and §6 says which sign each namespace can carry.

5 Typed overlap: the grading vector

5.1 One grade is not enough

The game note’s Proposition 7 is the lesson in miniature. There the connective was $|_0^{\mathcal{N}}$ and the concept was *fork*; grading over line names made the formula true of pairs of threats sharing a gap, which are not forks, and grading over square names made it exact. The choice of $\mathcal{N}$ was not a parameter of the encoding but the whole content of it.

At the level of learners the same demand recurs, and the namespace of a learner is not homogeneous. A shared metabolic channel and a shared probe channel are both “an overlap of one name” and they are not remotely the same relation.

Definition 11 (Typed namespaces). For a learner $\mathcal{L}$ with population P , the namespace $\mathcal{N}(\mathcal{L})$ decomposes into four sub-namespaces, each given by a name predicate:

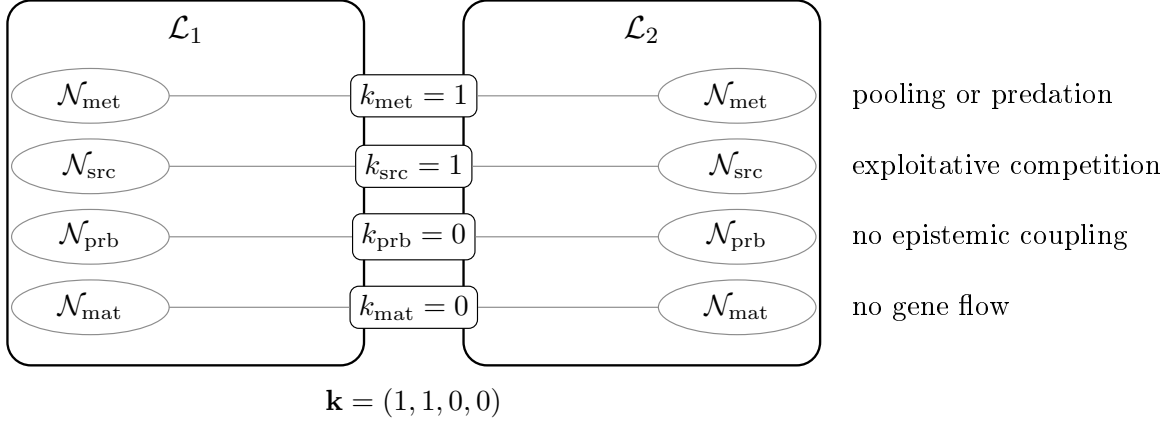


Figure 1: Two composed learners and their grading vector. A single scalar k would report “2” and say nothing; the typed vector says the two share a metabolic channel and a reservoir, cannot read each other’s assays, and cannot interbreed. Those are four independent dials and §5.2 reads the ecological relation off them.

- $\mathcal{N}_{\text{met}}(\mathcal{L})$, the *metabolic* names: the channels at which P ’s components hold their stacks, in the sense of [1, Def. 21];
- $\mathcal{N}_{\text{src}}(\mathcal{L})$, the *source* names: the channels of persistent emitters P draws on, the access profile of [1, Def. 39];
- $\mathcal{N}_{\text{prb}}(\mathcal{L})$, the *probe* names: the channels on which P ’s scientists install assays, which by [1, §12.6] are both-type channels supporting dialogue of depth at least two;
- $\mathcal{N}_{\text{mat}}(\mathcal{L})$, the *mating* names: the channels on which genomes are disclosed and recombined.

Definition 12 (Grading vector). For learners $\mathcal{L}_1, \mathcal{L}_2$ the grading vector is

$$\mathbf{k}(\mathcal{L}_1, \mathcal{L}_2) = (k_{\text{met}}, k_{\text{src}}, k_{\text{prb}}, k_{\text{mat}}), \quad k_{\tau} = |\text{fn}(P_1) \cap \text{fn}(P_2) \cap \mathcal{N}_{\tau}|,$$

the componentwise grade at which $P_1 \mid_{k_{\tau}}^{\mathcal{N}_{\tau}} P_2$ holds. Composition is *non-interacting* when $\mathbf{k} = \mathbf{0}$.

5.2 The relations, derived

The point of typing the overlap is that the classical table of ecological relations then falls out as sign conditions on the components, rather than being imported as a taxonomy. Write Δ_i for the change in learner i ’s expected residual life under composition relative to isolation.

Proposition 13 (Symbiosis is minimal metabolic overlap). *Let $\mathcal{L}_1 \mid \mathcal{L}_2$ have $k_{\text{met}} = 1$ with the shared name a pooling channel—a metabolic channel on which both parands both receive and reissue—and $k_{\tau} = 0$ otherwise. Then the composite is exactly the construction of [1, Prop. 44]: two specialists sharing a metabolic channel, each free to tune its access profile independently, with the yield pooled. Hence Proposition 43 there, which dominates the mixed strategy for an atom, does not apply.*

Remark 14 (The organism boundary is a grade). Read with Remark 40 of [1], Proposition 13 says the difference between “one dual-feeding organism” and “two specialists in symbiosis” is

overlap	relation	(Δ_1, Δ_2)	mechanism
$\mathbf{k} = \mathbf{0}$	neutralism	$(0, 0)$	Theorem 7: μ factors
$k_{\text{src}} > 0$ only	exploitative competition	$(-, -)$	dilution of a finite reserve (§10)
$k_{\text{met}} > 0$, asymmetric	predation	$(+, -)$	harvest as a receipt [1, Def. 23]
$k_{\text{met}} = 1$, pooled	mutualism (metabolic)	$(+, +)$	the composite of [1, Prop. 44]
$k_{\text{prb}} > 0$, one-way	commensalism (epistemic)	$(+, 0)$	reading a surface another paid for
$k_{\text{prb}} > 0$, two-way	mutualism (epistemic)	$(+, +)$	§6: formulae are not conserved
$k_{\text{prb}} > 0$, theft	exploitation (epistemic)	$(+, -)$	[1, §10]: holding a model exposes it
$k_{\text{mat}} > 0$	<i>not composition</i>	—	§8: the learners are one

Table 1: Ecological relations as sign conditions on the grading vector. Nothing in this table is posited; each row is a consequence already proved in [1] together with a choice of which $\mathcal{N}_\tau$ carries the overlap.

the value of k_{met} and where one chooses to draw the cut, not a fact about either. The endosymbiotic reading is available and we note it without leaning on it [15]: what the formalism contributes is that the boundary has a number attached to it.

Proposition 15 (The grading namespace must be typed). *Let $\mathcal{N} = \mathcal{N}_{\text{met}} \vee \mathcal{N}_{\text{src}} \vee \mathcal{N}_{\text{prb}} \vee \mathcal{N}_{\text{mat}}$ and suppose one grades only over $\mathcal{N}$. Then $|\mathcal{N}_k|$ cannot distinguish a composite in exploitative competition ($k_{\text{src}} = 1$, rest zero) from one in symbiosis ($k_{\text{met}} = 1$, rest zero), since both satisfy $|\mathcal{N}_1|$; and the two have opposite signs in Table 1. Hence the vector of Definition 12 is not a refinement of convenience.*

This is the same argument as [2, Prop. 7] one level up, and we take the recurrence as evidence that choosing the grading namespace is where the modelling work in this framework actually happens.

6 Cooperation and competition are structurally unlike

6.1 Two currencies, one conservation law

Under Remark 20 of [1] every metabolic transfer is zero-sum: predation is the purchase of somebody else’s remaining time. It follows immediately that cooperation cannot be a token transfer leaving both parties richer, and one might conclude that a conserved world has no room for mutualism at all. It does, and the room is in exactly one place.

Observation 16 (Tokens are conserved; formulae are not). Θ is fixed and never minted. A formula is a different kind of object: a genome is a quotation, $@P$ does not reduce, quotation is free while held, and inspection by a name predicate is grade-three access, the cheapest on the ladder. Nothing forbids two learners from holding the same formula, and one learner’s coming to hold it does not deprive the other. The ecology is conserved; the epistemics are not.

Proposition 17 (No metabolic mutualism). *Let $\mathcal{L}_1 \mid \mathcal{L}_2$ have $k_{\text{prb}} = k_{\text{mat}} = 0$, so that all overlap is metabolic or source. Then $\Delta_1 + \Delta_2 \leq 0$, with equality only when no shared name carries a transfer.*

Proof sketch. Residual life is stack divided by burn rate. Every step over a shared metabolic or source name either moves tokens from one parand to the other, which is zero-sum in stack and strictly negative in aggregate residual life whenever the two burn rates differ, or debits both parands for the rendezvous, which is negative. Pooling (Proposition 13) is the boundary case in which the transfer is a loop and the aggregate is unchanged; even there the rendezvous is metered, so equality requires no transfer at all. $\square$

So the only positive-sum overlap is epistemic, and it has two mechanisms, both already in the companion notes.

6.2 Cost relocation, and who is billed

Observation 18 (Cooperation is amortised perception). There are two routes to any predicate about an environment: *read* it, if the environment has computed it into legible structure, priced by formula depth; or *drive* it, by taking steps and watching, priced by rendezvous [2, §5.1]. If $\mathcal{L}_1$'s components compute a predicate into structure standing at a name in $\mathcal{N}_{\text{prb}}(\mathcal{L}_1) \cap \mathcal{N}_{\text{prb}}(\mathcal{L}_2)$, then $\mathcal{L}_2$ reads at grade one what it would otherwise drive at grade two. The total metering for both learners to know the predicate falls, and no token has been created.

Remark 24 of [2] makes the point that putting threats into the term relocates a cost from the scientist's budget to the world's, and that the interesting quantity is not whether the work is done but who is billed for it. Composition is what makes that question have two answers inside one system. A shared legible surface is stigmergy, and it is the only mechanism by which two mortal learners in a conserved world can both come out ahead.

Proposition 19 (Epistemic overlap can be strictly positive-sum). *Let φ be a predicate both learners require, costing κ_{drive} to establish by interaction and $\kappa_{\text{read}}(d) < \kappa_{\text{drive}}$ to read from structure, and let κ_{write} be the cost to $\mathcal{L}_1$ of computing it into structure at a shared probe name. Then $\Delta_1 + \Delta_2 > 0$ whenever*

$$2\kappa_{\text{drive}} > \kappa_{\text{drive}} + \kappa_{\text{write}} + \kappa_{\text{read}}(d),$$

that is, whenever writing once and reading once costs less than driving twice. This is not available over $\mathcal{N}_{\text{met}}$ or $\mathcal{N}_{\text{src}}$ by Proposition 17.

Remark 20 (And the same overlap can be theft). The mechanism cuts both ways and the notes already say so. Holding a hypothesis requires exposing it, because a guard is surface, so $\mathcal{N}_{\text{prb}}$ overlap that lets $\mathcal{L}_2$ read $\mathcal{L}_1$'s confirmed structure also lets it read $\mathcal{L}_1$'s theory without experimenting [1, §10]. Whether epistemic overlap is mutualism or exploitation is not settled by the grade; it is settled by whether the reading learner reciprocates, which is the plagiarism equilibrium left open as Problem 2 there. What the present framing adds is that the question is well posed as a sign condition on one component of $\mathbf{k}$.

Cooperation is epistemic; competition is metabolic.

7 Non-interference is not stable

Disjointness of namespaces is the condition under which composition is innocent, so it matters whether it can be relied upon. It cannot, by default, and the reason is one the companion note already records for a different purpose.

Proposition 21 (Disjointness is an observation, not a guarantee). *Let $\mathcal{L}_1 \mid \mathcal{L}_2$ have $\mathbf{k} = \mathbf{0}$ at some configuration. Because name manufacture is decentralised and freshness is scope-relative [1, Rem. 1], the two populations may subsequently manufacture structurally equal names at loci private to each. Hence $\mathbf{k} = \mathbf{0}$ at one configuration does not entail $\mathbf{k} = \mathbf{0}$ at a successor, and neither learner need have taken any step aimed at the other.*

Proof sketch. This is Remark 34 of [1]—closure is luck, not construction—together with Proposition 71 there, that decentralisation makes recombination generative precisely because two parents may independently hold structurally equal names. What is a source of novelty within a lineage is a source of unplanned coupling between learners. $\square$

Corollary 22 (There is no modular composition for free). *One cannot compose two learners and certify non-interference from the parts alone. Certification requires either a global check at each configuration, which is model checking and is priced, or a discipline on name manufacture.*

The discipline exists, it is cheap, and it is the one the board of [2, §4] already uses.

Proposition 23 (Rooting makes disjointness a theorem). *Let $\mathcal{L}_1$ and $\mathcal{L}_2$ manufacture every private name by quoting a term properly containing a distinguished root r_1 , respectively r_2 , with $r_1 \neq r_2$ and neither root occurring in the other's population. Then no name manufactured by $\mathcal{L}_1$ is structurally equal to a name manufactured by $\mathcal{L}_2$, and $\mathbf{k}$ is constant along every reduction that does not introduce a name from outside.*

Proof sketch. By Proposition 2 a manufactured name is $@C[r_i]$ for a context C with $C[r_i] \neq r_i$, so it codes a term properly containing r_i . Structural equality of $@C[r_1]$ and $@C'[r_2]$ would give $C[r_1] \equiv C'[r_2]$, and by well-foundedness of the coding—a name occurring in a term codes a strictly smaller term—this forces r_1 to occur in $C'[r_2]$, contradicting the hypothesis that r_1 does not occur in $\mathcal{L}_2$'s population. $\square$

Remark 24 (Modularity is purchasable, and the price is a naming discipline). This is the same trick as the board's three name families $@[*b, k]$, $@[*b, \text{"m"}, k]$, $@[*b, \text{"t"}, s]$, which are pairwise disjoint by construction and which is what makes grading over Th_b available in the first place. Applied to learners it says: root each learner's namespace at its own quotation and non-interference becomes a property one can rely on rather than one one must keep checking. It also says that a learner which *wants* to interact must import a name from outside its own root, which is a step, which is metered. Under rooting, every interaction has an identifiable first cause with a price attached—which is a better situation than the notes have had so far, where accidental coupling was possible and untraceable.

8 Merger

Overlap in the mating namespace is not a kind of interaction. It is the failure of the two learners to be two.

Proposition 25 (Mating overlap collapses the pair). *By [1, Prop. 53] two computations can recombine exactly at their homologous loci, so conspecificity is agreement of path sets and reproductive isolation is namespace disjointness. If $k_{\text{mat}} > 0$ and the shared names are homologous, then genomes cross the boundary, alleles from each appear in the other's descendants, and after finitely many generations neither population's namespace is described by its original predicate.*

Definition 26 (Merger). The merger $\mathcal{L}_1 \bowtie \mathcal{L}_2$ is the learner

$$(P_1 \mid P_2, \mathcal{N}_1 \vee \mathcal{N}_2, \mu_{\bowtie})$$

whose distribution ranges over configurations reachable under a recombination discipline admitting homologous loci *across* the two populations, together with the identification of the two mating namespaces.

Merger and composition agree on the first two components of Definition 3 and differ on the third, which is precisely why the third had to be part of the learner. As operators they behave differently in ways worth recording.

Proposition 27 (Merger is not composition). *Composition is associative and commutative up to $\equiv$ and admits the empty learner as a unit. Merger is commutative but is neither associative in general— $(\mathcal{L}_1 \bowtie \mathcal{L}_2) \bowtie \mathcal{L}_3$ and $\mathcal{L}_1 \bowtie (\mathcal{L}_2 \bowtie \mathcal{L}_3)$ admit different homology relations, since homology is not transitive across intermediate path sets—nor unital in the same sense, since $\mathcal{L} \bowtie \mathbf{0}$ imposes an identification that $\mathcal{L} \mid \mathbf{0}$ does not. Moreover merger is irreversible: no composition of the merged learner recovers the parts, because the alleles have crossed.*

Remark 28 (Why we want it separate). There is a temptation to treat merger as composition at high k_{mat} and be done. We resist it for two reasons. First, the sign structure of Table 1 has no entry for it: merger is not $(+, +)$ or $(+, -)$, because after it there are no two parties to assign signs to. Second, and more usefully, merger is the operator that changes the *identity* of a learner while composition changes only its situation, and §12 needs that distinction: a lineage escapes confinement by recombination *within* itself, and a federation escapes by composition *with* another. Merger is the third thing, and it escapes by ceasing to be the learner that was confined.

9 Phase is not compositional

Figure 2 of [1] divides the (ι, δ) plane into three regions: at low δ food lies in the open and senses suffice, so a theory is overhead; at high ι or high δ discovery costs more than it returns and everything starves; in between, hypothesis formation pays. The boundaries are where the foraging inequality changes sign.

Proposition 29 (Phase is not preserved). *There exist learners $\mathcal{L}_1, \mathcal{L}_2$ each of which lies in the science-pays band in isolation and whose composite lies outside it. Specifically:*

- (a) Downward, into foraging-suffices. *If $\mathcal{L}_2$'s components lie shallow to $\mathcal{L}_1$'s senses— δ small in the pair, which by Remark 3 of [2] means few plies of lookahead are needed to model them—then $\mathcal{L}_1$'s cheapest rung dominates its theory-holding rungs, since the prey is legible without inquiry.*
- (b) Downward, into starvation. *If composition raises ι past the upper boundary, which it does whenever $\mathcal{L}_2$ is stack-rich and reservoir-poor, then no available locus satisfies the foraging inequality for either party.*
- (c) Upward, out of starvation. *If $\mathcal{L}_1$ is below its minimum viable endowment and $k_{\text{met}} = 1$ is a pooling channel, then Proposition 13 rescues it.*

The interesting case is none of these three, because all three are couplings a modeller can see coming. The case worth measuring is the one in which the two learners never perceive each other at all.

10 Competitive dilution, measured

10.1 The model, and its one choice

Take $\mathbf{k} = (0, 1, 0, 0)$: two learners sharing exactly one source name and nothing else. Neither can perceive the other—no probe name is shared, so no assay of either reaches the other's surface—neither can harvest the other, and neither can interbreed with it. By Table 1 the relation is exploitative competition, and the only channel of influence is Θ itself.

The source is a single contract, hence rate-limited and non-exclusive [1, Def. 39]. We must say how the two learners' calls interleave, and this is the note's one genuinely free modelling choice.

Definition 30 (Access rate). A learner calling a source completes one assay per c metered units, where c is its expected metering per assay at the rung it holds. Learners queue at the source as fast as they complete assays, so learner i 's share of the source's payouts is

$$s_i = \frac{1/c_i}{\sum_j 1/c_j}.$$

The alternative—equal shares regardless of cost—is defensible if one reads the contract as serving callers round-robin rather than first-come. We work Definition 30 and note in §14 what changes under the alternative: the dilution survives, the asymmetry does not.

Proposition 31 (Competitive dilution). *Let two learners share one source of reserve R_0 paying quantum q per assay, at price p tokens per metered unit, with expected meterings c_1, c_2 per assay, both self-funding ($q > pc_i$). Then over the life of the source, learner i receives*

$$G_i = \frac{R_0}{q} \cdot \frac{c_j}{c_i + c_j} \quad (j \neq i)$$

assays and accumulates $\Sigma_i = G_i(q - pc_i)$. Consequently the number of assays of any deficit-running rung of cost $c^ > q/p$ that learner i can subsequently fund is*

$$D_i = \frac{\Sigma_i}{pc^* - q} = D_i^{\text{alone}} \cdot \frac{c_j}{c_i + c_j},$$

so that each learner retains exactly its own share of the reservoir as a fraction of the inquiry it could afford in isolation. The factor is independent of R_0 , q , p and c^ .*

Proof sketch. Immediate from Definition 30: composition scales G_i by s_i and everything downstream is linear in G_i . $\square$

Corollary 32 (Dilution is regressive). *A learner whose best affordable rung is self-funding loses assays but no depth, since the rung it holds needs no surplus. A learner still climbing—whose next rung runs a deficit and must be funded from accumulated surplus—loses depth in exact proportion to its share. Hence competition at a shared reservoir costs the unfinished science strictly more than the finished one, and costs nothing at all to a learner that has stopped.*

Remark 33 (Competition is a mechanism of confinement). Corollary 19 of [1] says poverty is confining: a scientist whose stack affords few evaluations per move takes the cheapest available move, the cheapest moves are the shortest, and by Proposition 13 short moves cannot leave their ball. Proposition 31 exhibits a mechanism of impoverishment that requires no contact whatever. A learner can be confined to a shallow theory by a competitor that cannot see it, cannot model it, and has no hypothesis about it—only a name in common. We think this is the most surprising consequence of the composition operator, and it is the classical competitive exclusion principle [13, 14] arriving in an epistemic currency rather than a demographic one.

11 The worked example: two games, one wall

11.1 A second environment

The game note ladderred noughts-and-crosses and found the complete theory dominated. To compose we need a second learner, and the second environment should differ in the one respect that matters: whether its complete theory is affordable.

rung	added rule	win	loss	cost	Δ win	return
<i>Nim(3, 4, 5)</i> <i>nim-sum = 2, an N-position; branching 12</i>						
0	$\top$	0.5000	0.5000	0.0	—	—
1	take-all	0.6567	0.3433	38.9	+0.1567	+0.00403
2	equalise	0.8899	0.1101	80.4	+0.2332	+0.00562
3	χ_{Nim}	0.9966	0.0034	99.6	+0.1067	+0.00555
<i>Nim(1, 2, 3)</i> <i>nim-sum = 0, a P-position; branching 6</i>						
0	$\top$	0.5000	0.5000	0.0	—	—
1	take-all	0.5917	0.4083	15.2	+0.0917	+0.00602
2	equalise	0.7958	0.2042	31.4	+0.2042	+0.01262
3	χ_{Nim}	0.9000	0.1000	38.5	+0.1042	+0.01476

Table 2: Two Nim ladders, exact. Contrast Table 1 of [2], where marginal return falls by more than two orders of magnitude along the ladder and the complete theory is dominated on both axes. Here return is flat in $\text{Nim}(3, 4, 5)$ and *rising* in $\text{Nim}(1, 2, 3)$, where the complete theory is the best purchase available.

We take Nim, normal play [11]. The ladder is built by the same method and priced by the same schedule $\kappa(d) = 2^d$, with the cost of a decision at rung r equal to $|M| \cdot \sum_{\text{rules } 1..r} \kappa(d_{\text{rule}})$ and $|M|$ the number of legal moves.

rung	formula added	kind and depth	κ
0	$\top$	—	0
1	$\exists s. \text{One}(s) \wedge \text{Takes}(s)$	spatial, depth 1	2
2	$\exists s. \text{Two} \wedge \text{Equalises}(s)$	spatial, depth 1	2
3	$\chi_{\text{Nim}} := \exists s. \text{NimSum}_0(s)$	name-level, depth 0	1

Rung 1 says: exactly one heap is nonempty, take it. Rung 2 says: exactly two heaps are nonempty, equalise them. Rung 3 is Bouton’s theorem: move to a position whose nim-sum is zero. All three are predicates on the heap sizes, read off the reflected structure of the position by a name predicate; none contains a modality, and rung 3 in particular requires no lookahead at all.

Observation 34 (The complete theory of Nim is name-level). χ_{Nim} is exact and contains no modality and no fixed point. This is Remark 11 of [1]—a finite theory is complete exactly when the environment’s regularity is structural—in its sharpest available form: Nim’s regularity is so structural that its complete theory sits at depth zero, where noughts-and-crosses needs $\nu X. \forall s. [K_s](\neg \text{Lost} \wedge \exists t. \langle\langle K_t \rangle\rangle X)$ and nine unfoldings.

11.2 The ladder pays, and here it keeps paying

All figures are exact over all plays, symmetrised over which side moves first, against a uniform-random opponent, computed by memoised recursion (Appendix B).

Observation 35 (Whether truth is affordable is a property of the environment). Three environments, three answers. In noughts-and-crosses χ costs 679.2, wins 0.873, and is strictly dominated by a name-level heuristic winning 0.918 at 255.9. In $\text{Nim}(3, 4, 5)$ χ_{Nim} costs 99.6, wins 0.9966, and is not dominated: it is more expensive than the rung below and strictly better. In $\text{Nim}(1, 2, 3)$ it is the best marginal purchase on the whole ladder, returning +0.0148 per metered unit where the rung below returns +0.0126 and the rung below that +0.0060. The returns *rise*.

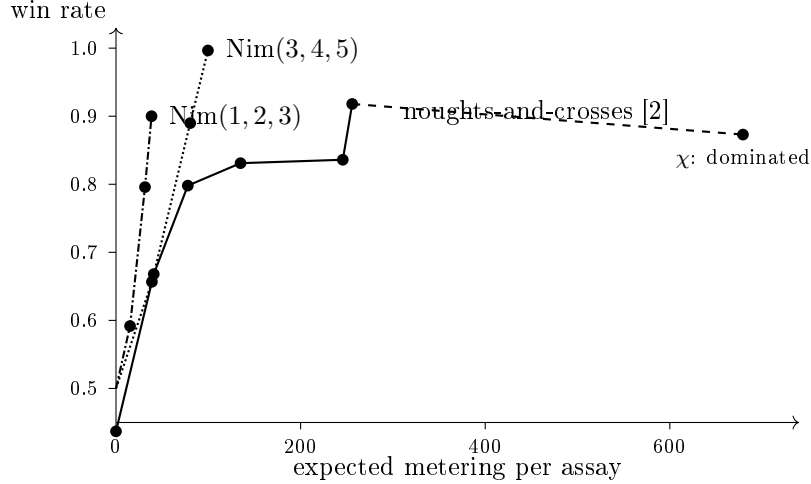


Figure 2: Three ladders on one pair of axes. The noughts-and-crosses curve turns down at its last rung—the complete theory costs more and wins less than the heuristic beneath it. Both Nim curves rise to the end, and the Nim(1, 2, 3) curve rises fastest at the top. Composition puts learners with curves of both shapes under one conserved Θ .

This is worth stating carefully because it is the hinge of the composition argument. Remark 17 of [1]—yield, not truth—says a lineage under selection drifts toward yield-biased search wherever yield and discrimination disagree. Table 2 shows they do not always disagree. Whether they do is fixed by the environment: by how much of its regularity is structural rather than reduction-following (Remark 11 there), and by the branching factor, which by Proposition 10 of [2] is the arity of the quantification over redex positions. So *is truth affordable?* is not a question about scientists. It is a question about worlds, and composition is the operator that puts two answers into one budget.

Remark 36 (χ in a lost position is a fall-through to $\top$). Nim(1, 2, 3) has nim-sum zero, so the first player has no move to a zero position and χ_{Nim} fires no rule; the policy falls through to uniform choice. It still wins 0.800 as first player because the random opponent errs and the complete theory punishes the error the moment it appears—but it has nothing to say about how to *provoke* one. As second player it wins 1.000. That the complete theory is silent on exploiting a fallible opponent is Observation 25 of [2] in a domain where χ is otherwise cheap and good, which is some evidence that the phenomenon is about completeness rather than about cost.

11.3 The composite

Let $\mathcal{L}_A$ be a population playing noughts-and-crosses and $\mathcal{L}_B$ a population playing Nim(1, 2, 3), each against the practice wall of [2, §3.2], rooted at distinct quotations so that Proposition 23 applies. They share exactly the wall: $\mathbf{k} = (0, 1, 0, 0)$. The wall’s reserve is $R_0 = 12,000$ at quantum $q = 6$, and the price is $p = 0.05$ tokens per metered unit, as in that note’s world.

At $p = 0.05$ and $q = 6$ a rung is self-funding exactly when its expected metering is below $q/p = 120$. For the noughts ladder that admits rungs 0, 1, 2 and excludes 3, 4, 5, 6; so $\mathcal{L}_A$ ’s deepest self-funding rung is rung 2, at $c_A = 77.8$ and a net of +2.11 tokens per assay, and every deeper rung must be funded out of accumulated surplus. For $\mathcal{L}_B$, χ_{Nim} costs $c_B = 38.5$, net +4.08: *the complete theory of $\mathcal{L}_B$ ’s environment is self-funding, and the incomplete theory of $\mathcal{L}_A$ ’s environment is not.*

	share of wall	assays	terminal surplus	rung-5 assays funded
$\mathcal{L}_A$ alone	1.000	2000	4220	621
$\mathcal{L}_A$ composed	0.331	662	1397	206
$\mathcal{L}_B$ composed	0.669	1338	5452	— (none needed)

Table 3: The composite, exactly. $\mathcal{L}_A$ retains 0.331 of the deep inquiry it could afford alone—identically its share of the wall, per Proposition 31—and loses 66.9%. $\mathcal{L}_B$ loses 33.1% of its assays and none of its depth, because the rung it holds is self-funding. Neither learner perceives, models, or touches the other.

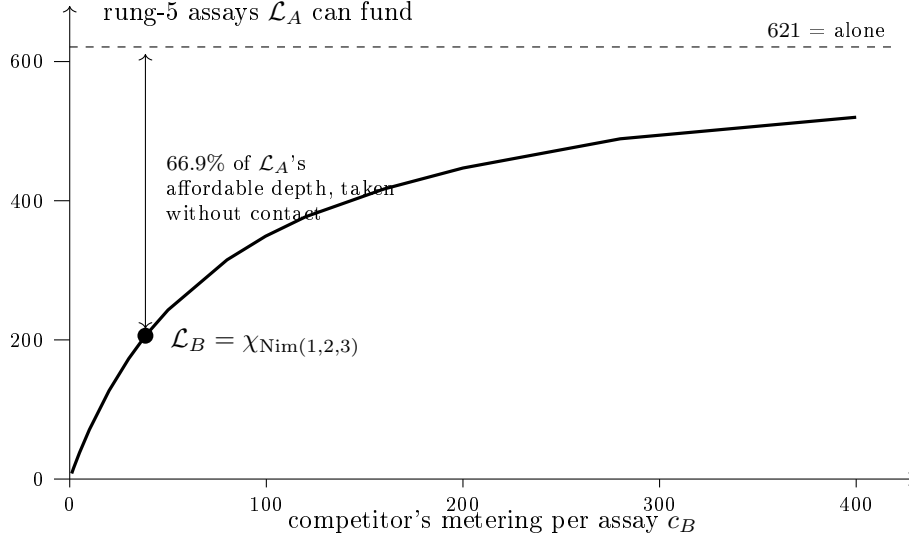


Figure 3: Competitive dilution. The curve is $D_A(c_B) = D_A^{\text{alone}} \cdot c_B / (c_A + c_B)$ with $c_A = 77.8$, exactly as Proposition 31 gives it. As the competitor’s science gets cheaper, the affordable depth of the learner whose science is still unfinished goes to zero.

Observation 37 (The measurement). $\mathcal{L}_A$ ’s affordable depth falls from 621 assays of rung-5 play to 206. The mechanism is entirely Θ : $\mathcal{L}_B$ holds no hypothesis about $\mathcal{L}_A$, has no probe name in common with it, could not derive its metabolic channel if it wanted to, and would gain nothing by doing so. It simply finishes its assays faster, because its environment admitted a cheap complete theory, and therefore reaches the wall more often.

Observation 38 (And it is regressive). $\mathcal{L}_B$ is unaffected in depth. Its best rung is self-funding, so the surplus it forgoes buys it nothing it needs. $\mathcal{L}_A$ is affected in depth alone, because everything it still wants to learn runs a deficit. The learner with the finished science pays in currency it does not spend; the learner with the unfinished science pays in the only currency that matters to it.

11.4 Succession is a change of the grading vector

The wall is finite, so this composite has an ending, and the ending is where the typing of the overlap earns its keep.

Proposition 39 (Exhaustion forces a change of type). *Let $\mathcal{L}_1 \mid \mathcal{L}_2$ have $\mathbf{k} = (0, 1, 0, 0)$ with the shared source exhaustible. When the source is exhausted, the foraging inequality fails at every source locus for both learners. The composite then either dies or acquires $k_{\text{met}} > 0$*

or $k_{\text{prb}} > 0$; that is, succession in the sense of [1, Prop. 45] is precisely a change in which component of the grading vector is nonzero.

Corollary 40 (The change is priced, and priced asymmetrically). *Acquiring $k_{\text{met}} > 0$ requires deriving a name in the other learner’s metabolic namespace, which under capability gating is real work: a hypothesis describing that namespace must be formed and installed as a guard. By Proposition 18 of [1] the branching a learner can survey is its stack divided by the cost of one check. Hence the transition is available to whichever learner retains apparatus for forming namespace hypotheses about adaptive subjects—which by Corollary 42 there is the prey-feeder, and in this composite is the learner whose science was unfinished.*

In the worked composite, $\mathcal{L}_B$ ends the source phase holding 5452 tokens and $\mathcal{L}_A$ holds 1397; $\mathcal{L}_B$ ’s stack is 3.90 times $\mathcal{L}_A$ ’s. By the foraging inequality that stack is worth opening at any concealment depth these prices make affordable. So the learner that was confined during the source phase is the one positioned to harvest at succession, and the capital it harvests was accumulated by the learner that confined it.

Remark 41 (Offered as a structure, not a moral). We resist reading anything edifying into this. What the framework says is narrow and checkable: a cheap complete theory is a source-like access profile—non-adaptive, converged, with no standing revision cost—and by Corollary 42 of [1] the apparatus for modelling adaptive subjects is a heterotroph’s expense, not carried by anything that does not need it. A learner that has finished its science has, by construction, stopped paying for the machinery that would let it form a hypothesis about a competitor. That is not a punishment for being right. It is what being right in a closed domain costs, and it is the same fact as Proposition 32 of [2]: a solved science is a dead ecology.

12 The three units of learning

Collecting the three notes, the units of learning are distinguished by exactly one thing: the revision distance each can reach.

unit	mechanism	reachable move	authority
individual	walk the relaxation lattice	confined below 2^{-n}	[1, Prop. 13]
lineage	crossover at a shallow locus	2^{-1}	[1, §12.10], [2, §9.3]
composed learner	adopt another root	the whole tree	§4, Prop. 23

Proposition 42 (Composition reaches what no lineage reaches). *A lineage’s recombination acts at loci of its own genomes, so every allele it can install already occurs in one of its members and every namespace it can come to describe is generated from its own root. Under Proposition 23 a change of root is therefore unreachable by recombination. It is reachable by composition, since $\mathcal{N}_1 \vee \mathcal{N}_2$ contains names manufactured from both roots and the composite may install guards over either.*

Remark 43 (The series, and where it stops). The game note measured the middle row: the corner-opening allele was unreachable to any individual by deepening, free to a lineage by crossover, and the invasion was unconditional. The top row is the analogue one step up, and its content is that whatever a lineage’s early commitments fixed—not merely its opening, but the family of names its whole hypothesis language is generated from—is fixed for good against every mechanism internal to it. Composition is the only operator in these notes that changes it, and merger (§8) is the only one that changes the learner’s identity along with it. We do not know whether the series continues, and Problem 6 asks.

13 Composition is the exit from the dead ecology

Proposition 32 of [2] says a population all of whose members hold theories that draw against one another performs no harvest, burns tokens to live, and starves; and that the better its science the sooner it dies. Remark 34 there lists three exits and takes none of them: enlarge the game so χ is unaffordable, let the rules drift, or admit an external source.

The first exit is composition.

Proposition 44 (Composition restores harvest). *Let $\mathcal{L}$ be a learner in the terminal state of [2, Prop. 32]: harvest rate zero, χ held by every member, σ declining monotonically. Let $\mathcal{L}'$ be any learner whose probe namespace is generated from a distinct root, so that $\mathcal{L}$ holds no hypothesis satisfied by $\mathcal{L}'$'s components. Then in $\mathcal{L} \mid \mathcal{L}'$ with $k_{\text{met}} > 0$ the foraging inequality is satisfiable again, because $\mathcal{L}'$'s stacks are concealed at a depth $\mathcal{L}$ has not paid to see, and paying is once more a purchase with a return.*

Remark 45 (Enlarging the game and composing are the same operation). This is the content we would most like a reader to take away. “Enlarge the game so that the complete theory is unaffordable” sounds like a modelling intervention—a change of domain imposed from outside. Under Definition 5 it is an operation *inside* the framework, performed by the ecology on itself, requiring no new rewrite rule and no new sort. A science that has closed reopens by meeting a science that has not. The heat death of [2] was never a property of noughts-and-crosses; it was a property of a world with one game in it.

14 What this shows, and what it does not

Shown. That a learner may be taken to be a weighted population and hence a namespace, and that under that reading composition is parallel composition on terms, disjunction on namespaces, and undetermined on distributions (§3–§4). That disjointness of namespaces is exactly factorisation of the distribution, so the lax comparison map’s defect *is* the interaction and is a quantity rather than a caveat (Theorem 7, Proposition 9). That a typed grading vector recovers the ecological relations as sign conditions (§5). That conservation makes epistemic overlap the only positive-sum overlap (§6). That non-interference is unstable by default and purchasable by rooting (§7). That phase is not compositional (§9) and that competitive dilution confines a learner without contact, in a closed form and on exact numbers (§10–§11).

Not shown. We have not proved a monoidal-category theorem: Proposition 9 gives the comparison map and its invertibility condition, and the coherence conditions, the unit laws up to the identifications merger imposes, and the relationship to the fibration’s section are left open. Definition 30 is a modelling choice; under the round-robin alternative each learner receives half the payouts and the retained fraction becomes 1/2 for both, so the *dilution* survives and the *regressiveness* weakens to whatever asymmetry the rungs themselves carry—in the worked composite $\mathcal{L}_A$ would retain 0.500 rather than 0.331, and still lose depth where $\mathcal{L}_B$ loses none. As in [2] the measurements are exact computations over policies rather than observations of a running population: the ecology of Appendix A is written and not run to a fixed point, so every claim about invasion and succession is a claim about payoffs. And the Nim ladders, like the noughts ladder, are priced by a schedule that is a choice.

The honest summary. The composition operator was, going in, an obvious formality—parallel composition is already there, and saying that learners compose by it looks like a restatement. What the exercise produced instead was three things we did not put in: that

the probabilistic content of composition is exactly a graded separation, that a typed grading vector is forced rather than convenient, and that two learners which cannot perceive each other can nonetheless confine each other's inquiry by a measurable amount.

15 Open problems

1. **Coherence.** Is $(\text{Eco}, |, \mathbf{0})$ lax monoidal with λ of Proposition 9 as the structure map, and is the canonical section of [1, Prop. 80] lax monoidal over it? The obstruction to strictness is located; the coherence conditions are not checked.
2. **Is the grading vector complete?** Definition 11 gives four sub-namespaces because four are what the companion notes distinguish. Is there a fifth? The candidate is the *arena* namespace: the names by which an admitted context class $\mathcal{A}$ is realised ([2, Obs. 4]). Two learners sharing an arena but no other name play the same game against different opponents, which is neither of the relations in Table 1.
3. **Optimal k .** Given a distribution over environments, is there an optimal grading vector for a learner to hold—an amount of overlap that maximises expected residual life? §6 says the $\mathcal{N}_{\text{prb}}$ component should be positive and the $\mathcal{N}_{\text{met}}$ component should be zero or pooled, but the trade-off against the exposure of Lemma 33 of [1] is not worked.
4. **Run the composite.** Appendix A specifies a two-learner world; §11 computes payoffs. Sampling trajectories by the Gillespie construction of [7] would turn Table 3 from a payoff calculation into an ensemble result, and would settle whether the succession of Proposition 39 actually occurs or whether $\mathcal{L}_A$ starves first.
5. **The plagiarism equilibrium, now well posed.** Problem 2 of [1] asked whether honest science survives theory theft. With the grading vector it becomes: for which k_{prb} is the two-way reading equilibrium stable against a one-way reader, and does the answer depend on the shape of the ladder (Observation 35) in each learner's environment?
6. **Does the series continue?** Individual, lineage, composite: each reaches a revision the one below cannot. Is there a fourth unit, and if so what does it reach? The natural candidate is a distribution over *composites*—a distribution over grading vectors rather than over configurations—and we do not know whether that is a new object or the same one at a coarser resolution.
7. **Cheapness versus completeness.** Observation 35 shows that whether χ is a good purchase varies by environment. Is there a criterion—presumably in terms of Remark 11 of [1], how much of the environment's regularity is structural, together with the branching factor—that predicts the shape of the ladder without computing it? For a composite this would predict who dilutes whom in advance of any measurement.

A A composite world, in rholang

The listing below is the composite of §11: two learners rooted at distinct quotations, sharing exactly one source name, with the typed namespaces of Definition 11 written as name predicates and the grading vector as a family of graded separations over them. Guards using a structural or modal connective are marked `//! ideal` and specify a check rather than perform one; everything else runs. Conservation is auditable by inspection: `Wall`, `Harvest` and `Beget` move tokens, `Live` debits them, and nothing mints.

```

// =====
// TWO LEARNERS, ONE WALL
// =====
//
// Companion listing to "Composing Learners". The world is literally
//
// Wall | L_A | L_B
//
// and every claim of the note is a claim about what that bar does.
//
// ROOTING (Prop. "Rooting makes disjointness a theorem"). Each learner
// manufactures every private name by quoting a term containing its OWN root.
// L_A's names are @[*rootA, ...] and L_B's are @[*rootB, ...]. By freshness-
// by-quotation no name of one is structurally equal to a name of the other,
// so  $k = 0$  holds on every component of the grading vector EXCEPT the wall,
// whose name neither learner manufactured and both were handed. That single
// imported name is the whole of the coupling.
//
// TYPED NAMESPACES (Def. "Typed namespaces"). Four predicates per learner:
//
// Met(r) := @[ *r, "m", _ ] metabolic: where stacks rest
// Src(r) := @[ *r, "s", _ ] source: emitters drawn upon
// Prb(r) := @[ *r, "p", _ ] probe: where assays are installed
// Mat(r) := @[ *r, "g", _ ] mating: where genomes are disclosed
//
// The grading vector of the composite is then the four-tuple
//
// k_tau = | fn(L_A) /\ fn(L_B) /\ N_tau |
//
// and the note's claim is that this world has  $k = (0,1,0,0)$ .

new stdout('rho:io:stdout'), display, wall, rootA, rootB,
  Live, Wall, Assay, Ladder, Body, Harvest, Beget, Merge, Learner
in {

  for (@line <= display) { stdout!(line) }

  // =====
  // METABOLISM. Reclaim - debit - reissue. Between the receipt and the send
  // the stack is a message in flight, which is what Harvest exploits.
  // =====
  | contract Live(m, @burn, step) = {
    for (@sigma <- m) {
      match sigma > burn {
        true => { m!(sigma - burn) | *step | Live!(*m, burn, *step) }
        false => { display!("DIED", "could not fund a step") }
      }
    }
  }

  // =====
  // THE SHARED SOURCE. One contract, hence rate-limited; non-exclusive, since
  // anyone holding the name may call it; non-adaptive; and finite.
  //
  // This contract IS the entire overlap. Note what it does NOT do: it does
  // not know who is calling, does not adapt to callers, and cannot be used by
  // one caller to observe another. Competitive dilution is delivered by a
  // process with no model of anybody.
  // =====
  | contract Wall(w, @quantum, @reserve) = {
    for (ret <- w) {
      match reserve >= quantum {

```



```

[h ...t] => {
  Body!(*r, which, rung, burn, h)
  | Learner!(*r, which, rung, burn, t)
}
}
}

| contract Body(r, @which, @rung, @burn, @i) = {
  new hyp, score in {
    @[*r, "m", i]!(400) // the stack: in Met(r)
    | score!(0, 0)
    | Ladder!(which, rung, *hyp)
    | Live!(@[*r, "m", i], burn, @{
      new ret in {
        for (@h <- hyp) { hyp!(h) | Assay!(*r, i, h, *ret) }
        | for (@o <- ret) {
          for (@c, @f <- score) {
            match o {
              ("confirm", q) => { score!(c + 1, f)
                                | for (@s <- @[*r, "m", i]) {
                                  @[*r, "m", i]!(s + q) } }
              _ => { score!(c, f + 1) }
            }
          }
        }
      }
    })
  }
}

// =====
// SUCCESSION. When the wall returns 0 forever, the only credit left is in
// another learner's Met namespace -- which is a namespace this learner has
// never described, because k_prb = 0 kept it invisible. Acquiring it is a
// step, and the step is metered: a hypothesis over Met(rOther) must be
// formed and installed. That is the change of grading vector, and it is
// available only to a learner that still carries the apparatus.
// =====
| contract Harvest(@rSelf, @i, @rOther, ret) = {
  //! ideal -- the guard is a name predicate over the OTHER root's
  // metabolic namespace. Forming it is what costs; firing it is a comm.
  for (@mPrey <- @[*rSelf, "p", "hunt"]
      where mPrey != @{ Met(rOther) }) {
    for (@sigma <- @{mPrey}) {
      for (@own <- @[*rSelf, "m", i]) {
        @[*rSelf, "m", i]!(own + sigma)
        | ret!(("harvest", sigma))
        | display!(("HARVEST across roots", sigma))
      }
    }
  }
}

| contract Beget(m, @alleles, @endow, ret) = {
  for (@sigma <- m where sigma > endow + 40) {
    m!(sigma - endow)
    | match alleles {
      [w, rg] => {
        new childM, code in {
          code!( @{ Body!(*childM, w, rg, 2 + rg, endow) } ) // inert
          | for (g <- code) { *g | childM!(endow) } // metered
          | ret!(("born", w, rg))
        }
      }
    }
  }
}

```

```

    }
  }
}
}

// =====
// MERGER -- a DIFFERENT OPERATOR, and this is the point of the section.
//
// Composition is the bar. It is free, it is associative and commutative,
// and it leaves both learners identifiable afterwards. Merger identifies
// the two MATING namespaces, which is a step, is metered, and is
// irreversible: after it there is no composition recovering the parts,
// because alleles have crossed roots.
//
// Note that Merge is a CONTRACT and '|' is not. That asymmetry is the
// formal content of "merger is a separate operator".
// =====
| contract Merge(@rA, @rB, ret) = {
  new rC in {
    // the merged learner's mating namespace admits homologous loci
    // across BOTH roots; this identification is what cannot be undone
    @[*rC, "g", "admits"]!([rA, rB])
  | ret!(*rC)
  }
}

// =====
// THE WORLD. Theta = 12000 (wall) + 3 x 400 (L_A) + 3 x 400 (L_B) = 14400.
//
// The composite is the bar between the two Learner calls. Delete either and
// the other's numbers change -- by exactly the factor of Prop. "competitive
// dilution", and by nothing else, since no other name is shared.
// =====
| Wall!(*wall, 6, 12000)

| Learner!(*rootA, "ttt", 2, 4, [0, 1, 2]) // noughts: rung 2, c = 77.8
| Learner!(*rootB, "nim", 3, 5, [0, 1, 2]) // nim: chi_Nim, c = 38.5

// -----
// WHAT TO WATCH
//
// * The overlap is ONE NAME. 'wall' is the only identifier appearing in
// both Learner subtrees. Everything else is manufactured under a root.
// k = (0,1,0,0), and it stays there by construction, not by luck.
//
// * Nobody models anybody. No guard anywhere mentions the other root.
// L_B has no hypothesis about L_A and would gain nothing from one. The
// dilution happens anyway.
//
// * L_B finishes and L_A does not. L_B holds the complete theory of its
// environment at burn 5; L_A holds an incomplete one at burn 4 and cannot
// afford the rungs it still wants. The one still climbing is the one the
// competition costs.
//
// * And it ends. When the wall returns 0, Assay's confirm guard stops
// crediting, Live keeps debiting, and the only remaining credit is inside
// the other root's Met namespace. Reaching it is Harvest, and Harvest's
// guard is a hypothesis somebody has to be able to afford to form.
// -----
}

```

B Reproducing the numbers

All figures in §11 are exact computations, not samples. The noughts-and-crosses ladder figures—win rates 0.437, 0.668, 0.798, 0.831, 0.836, 0.918, 0.873 and expected meterings 0.0, 40.9, 77.8, 134.9, 245.7, 255.9, 679.2—are quoted from Table 1 of [2] and were not recomputed here.

The Nim ladders are computed by `compose.py`, which enumerates positions as sorted tuples of heap sizes, evaluates each rung by memoised recursion over all plays with uniform tie-breaking within a rung and a uniform-random opponent, and charges $|M| \cdot \sum_{1..r} \kappa$ at each decision of the policy player with $\kappa(d) = 2^d$ and depths 1, 1, 0 for the three rules. Probabilities are carried as exact rationals and converted to decimal only for display. Figures are symmetrised over which side moves first; the unsymmetrised figures are reported separately where the colour asymmetry is the point.

The composite of §11 uses $R_0 = 12,000$, $q = 6$, $p = 0.05$, shares by Definition 30, and the identity of Proposition 31; the computation is a few lines and is included in the same script for auditability rather than for difficulty.

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